Physics Part 5

What is Doppler Effect

It is the variation of the pitch heard from a moving source of sound or by a moving observer. Named after Austrian Physicist, Christian Andreas Doppler.

As the source approaches an observer, the frequency or pitch of the wave will appear higher than it actually is. As the source moves away from the observer, the frequency or pitch of the wave will appear lower than it really is.

This is sometimes known as Doppler shift.

Relation between apparent frequency and the frequency of the source is expressed as:

$$f_o = \frac{V - v_o}{V - v_s} f_s$$

where:
fo - frequency of the observer
fs - frequency of the source
V - speed of wave relative to medium
vs - speed of source
vo - speed of observer

LIGHT

Light is to the aspect of radiant energy of which an observer is visually aware. The range of wavelength that is visible as light is from 3900 A to 7600 A. The speed of light in vacuum is approximately 3.00 x 10^8 m/s.

PROPERTIES OF LIGHT

Luminous flux is the amount of visible radiation passing per unit time. The SI is lumen. **Lumen** is defined as the flux emitted by a point source of 1 candela through a solid angle of 1 steradian.

$$I = \frac{F}{\omega}$$

where:
I – luminous intensity (candela)
F – luminous flux (lumen)
ω – solid angle (steradian)

The total luminous flux emitted by a point source is:

$$F = 4\pi I$$

Illuminance is the luminous flux per unit area. Its unit is **lumens per square foot** or lumens per square meter. Illuminance is sometimes referred to as **illumination**.

$$E = \frac{F}{A}$$

where:
E – illuminance (lumen/m^2)
F – luminous flux (lumen)
A – area (m^2)

For light from a point source, the illuminance on the surface is given by the inverse-square law, as follows:

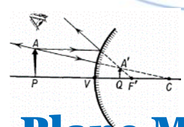
$$E = \frac{1}{s^2} \cos\theta$$

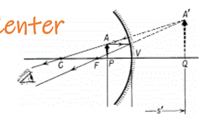
Luminance is the luminous intensity per unit projected area emitted by an extender source.

$$B = \frac{I}{A}$$

where:
B – luminance (candela/m^2)
I – luminous intensity (candela)
A – area (m^2)

Mirrors and Lenses





Plane Mirrors

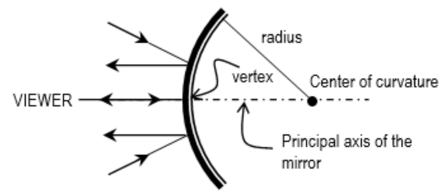
Plane mirror is a mirror with a flat or planar reflective surface. This is the simplest mirror around. The flat surfaces reflect a beam of light in one direction instead on either scattering it widely in many directions or absorbing it. The angle of reflection of this mirror is equal to the angle of incidence

Properties of Plane Mirrors:

1. Plane mirrors produce virtual images.
2. Plane mirrors produce upright images.
3. The presence of the left-right reversal phenomena.
4. Images produced are of the same size as object.
5. Images produced are with the same distance as object.

Spherical Mirrors

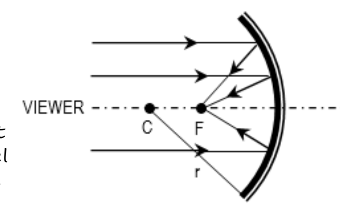
A **spherical mirror** is a mirror which has the shape of a piece cut out of a spherical surface. In a spherical mirror, the reflecting surface is a section of a sphere. There are two types of spherical mirrors, namely concave and convex. A **convex mirror** curves away from the viewer with its center of curvature is behind the mirror. A convex mirror is also called a **diverging mirror** since the reflection of a set of parallel rays is a set of diverging rays. The focal point of a convex mirror is on the principal axis a distance 0.5r behind the mirror where r is the radius of the sphere.



Convex mirror

Principal rays for convex mirrors:
1. A ray parallel to the principal axis is reflected as if it comes from a focal point.
2. A ray along a radius is reflected back upon itself.
3. A ray directed toward the focal point is reflected parallel to the principal axis.
4. A ray incident on the vertex of the mirror reflects at an equal angle to the axis.

A **concave mirror** curves toward the viewer with its center of curvature in front of the mirror. A concave mirror is also called **converging mirror** since it makes parallel rays converge to a point.



Concave mirror

Principal rays for concave mirrors:
1. A ray parallel to the principal axis is reflected through the focal point.
2. A ray along a radius is reflected back upon itself.
3. A ray along the direction from the focal point to the mirror is reflected parallel to the principal axis.
4. A ray incident on the vertex of the mirror reflects at an equal angle to the axis.

The mirror equation: $\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$ $f = \frac{r}{2}$

Magnification: $M = \frac{\text{image size}}{\text{object size}} = -\frac{q}{p}$

where:
p = object distance f = focal length
q = image distance r = radius of curvature

LIGHT SPECTRUM